

Enterprise Network Infrastructure

--Mega lab--

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1. Introduction

This project represents the design and implementation of a company's internal network.

The main goals of the project are:

- Dividing the network into departments using VLANs and Subnetting.
 - Providing dynamic IP addressing using DHCP.
 - Enabling inter-VLAN communication via Router-on-a-Stick.
 - Applying Layer 2 security features to protect the network.
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2. Network Topology

The network consists of:

- One Router (acting as default gateway and DHCP server).
- One Main Switch connected to the router.
- Multiple Access Switches connected to the main switch.
- End Devices (PCs, Printers, Servers) distributed across different departments.

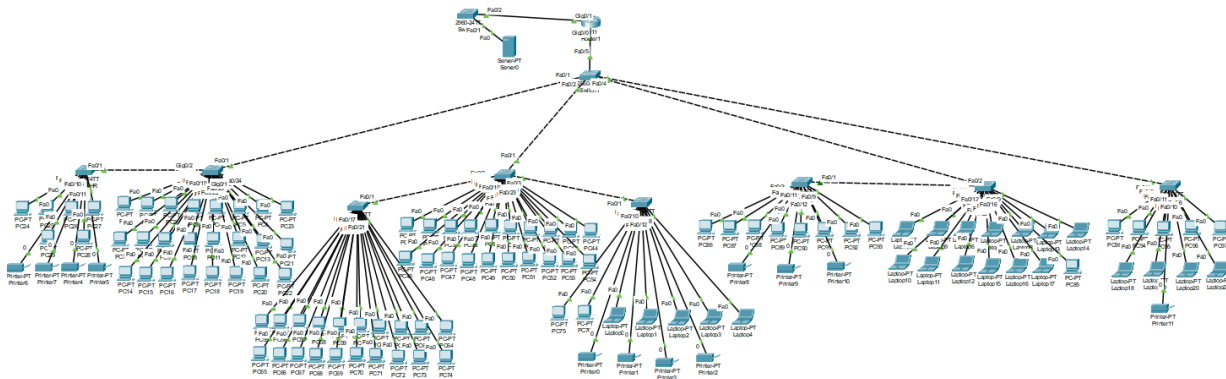


Figure 1: Network Topology

3. Subnetting Design Explanation

The company's main network 192.168.1.0/24 has been divided into multiple subnets to meet the specific needs of each department, with the following goals:

- Improve IP address management
- Reduce broadcast domains
- Increase overall network security
- Allocate IP addresses based on the actual number of devices in each department

◆ 1. IT Department

- Subnet ID: 192.168.1.0
- IP Range: 192.168.1.1 – 192.168.1.62
- Broadcast Address: 192.168.1.63
- Subnet Mask: 255.255.255.192 (/26)
- Available Hosts: 62 devices

The largest subnet is assigned to the IT department since it has the highest number of devices and servers.

◆ 2. HR Department

- Subnet ID: 192.168.1.64
- IP Range: 192.168.1.65 – 192.168.1.126
- Broadcast Address: 192.168.1.127
- Subnet Mask: 255.255.255.192 (/26)
- Available Hosts: 62 devices

The HR department requires a mid-sized network, so it was given the same subnet size as IT but with a different range.

◆ 3. Sales Department

- Subnet ID: 192.168.1.128
- IP Range: 192.168.1.129 – 192.168.1.158
- Broadcast Address: 192.168.1.159
- Subnet Mask: 255.255.255.224 (/27)
- Available Hosts: 30 devices

The Sales department is smaller, so a smaller subnet was allocated with a suitable number of addresses.

◆ 4. Accounting Department

- Subnet ID: 192.168.1.160
- IP Range: 192.168.1.161 – 192.168.1.174
- Broadcast Address: 192.168.1.175
- Subnet Mask: 255.255.255.240 (/28)
- Available Hosts: 14 devices

The Accounting department has the fewest devices, so the smallest subnet was assigned to minimize address waste.

Conclusion

- The main company network has been efficiently divided into four subnets, each tailored to the number of devices in its respective department.
- Variable Length Subnet Masking (VLSM) was used to optimize address space usage and minimize waste.
- The current design allows for future scalability, enabling the addition of new departments within the same /24 address block

4. IP Addressing Scheme

Department	Network Address	Subnet Mask	CIDR	Usable Range	Default Gateway
IT	192.168.1.0	255.255.255.192	/26	192.168.1.1 – 192.168.1.62	192.168.1.1
HR	192.168.1.64	255.255.255.192	/26	192.168.1.65 – 192.168.1.126	192.168.1.65
Sales	192.168.1.128	255.255.255.224	/27	192.168.1.129 – 192.168.1.158	192.168.1.129
Accounting	192.168.1.160	255.255.255.240	/28	192.168.1.161 – 192.168.1.174	192.168.1.161

VLANs Configuration

- **VLAN 10** – IT Department
- **VLAN 20** – HR Department
- **VLAN 30** – Sales Department
- **VLAN 40** – Accounting Department

Each VLAN is associated with its own subnet. The switches have been configured with **Access Ports** for end devices and a **Trunk Port** for communication with the router (using the **Router-on-a-Stick** method).

```
Switch# show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                           Gig0/2
10   Accounting              active
20   sales                   active
30   IT                      active
40   HR                      active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
Switch#

Switch#show int trunk

Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     auto      n-802.1q       trunking    1
Fa0/2     auto      n-802.1q       trunking    1
Fa0/3     auto      n-802.1q       trunking    1
Fa0/4     on        802.1q         trunking    1
Fa0/5     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
Fa0/4     1-1005
Fa0/5     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40
Fa0/2     1,10,20,30,40
Fa0/3     1,10,20,30,40
Fa0/4     1,10,20,30,40
Fa0/5     1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
```

Figure 2:VLANs and Trunk Configuration

5. Inter-VLAN Routing (Router-on-a-Stick)

To enable communication between VLANs, Router-on-a-Stick was configured:

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.1 255.255.255.192
```

```
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.1.65 255.255.255.192
```

```
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.1.129 255.255.255.224
```

```
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.1.161 255.255.255.240
```

- Each VLAN has its own default gateway.
- Devices can communicate across VLANs via the router.

6. DHCP Configuration

Dynamic IP assignment was provided using the router's built-in DHCP service.

```
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp excluded-address 192.168.1.65 192.168.1.70
ip dhcp excluded-address 192.168.1.129 192.168.1.130
ip dhcp excluded-address 192.168.1.161 192.168.1.162
```

```
ip dhcp pool IT
network 192.168.1.0 255.255.255.192
default-router 192.168.1.1
dns-server 192.168.1.200
```

```
ip dhcp pool HR
network 192.168.1.64 255.255.255.192
default-router 192.168.1.65
dns-server 192.168.1.200
```

```
ip dhcp pool Sales
network 192.168.1.128 255.255.255.224
default-router 192.168.1.129
dns-server 192.168.1.200
```

```
ip dhcp pool Accounting
network 192.168.1.160 255.255.255.240
default-router 192.168.1.161
dns-server 192.168.1.200
```

- Reserved addresses for gateways and servers.
- Each department receives IPs dynamically within its subnet.
- A local DNS server (192.168.1.200) was used.

```

-----
R1# show running-config
Building configuration...

Current configuration : 1757 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp excluded-address 192.168.1.65 192.168.1.70
ip dhcp excluded-address 192.168.1.129 192.168.1.130
ip dhcp excluded-address 192.168.1.161 192.168.1.162
!
ip dhcp pool IT
network 192.168.1.0 255.255.255.192
default-router 192.168.1.1
dns-server 192.168.1.200
ip dhcp pool HR
network 192.168.1.64 255.255.255.192
default-router 192.168.1.65
dns-server 192.168.1.200
ip dhcp pool Sales
network 192.168.1.128 255.255.255.224
default-router 192.168.1.129
dns-server 192.168.1.200
ip dhcp pool Accounting
network 192.168.1.160 255.255.255.240
default-router 192.168.1.161
dns-server 192.168.1.200
.

interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.1.161 255.255.255.240
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.1.129 255.255.255.224
!
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 192.168.1.1 255.255.255.192
!
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 192.168.1.65 255.255.255.192
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
!
interface Vlan1
no ip address
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^CWARNING: Unauthorized access is prohibited! ^C

```

Figure 3:DHCP Pools Configuration

7. Security Measures Implemented

Port Security

- Prevents unauthorized device connections.
- Configured with sticky MAC, maximum 2 devices per port.

DHCP Snooping

- Blocks rogue DHCP servers.
- Trusted only uplinks to router.

Dynamic ARP Inspection (DAI)

- Protects against ARP spoofing.
- Uses DHCP Snooping database for validation.

BPDU Guard with PortFast

- Protects spanning-tree by disabling ports receiving unauthorized BPDUs.
- Applied on all access ports.

8. Conclusion

The company's network was successfully designed and implemented using:

- Subnetting to divide 192.168.1.0/24 into departmental networks.
- VLANs and Router-on-a-Stick for inter-VLAN routing.
- DHCP for dynamic addressing.
- Layer 2 security features to protect against common LAN attacks.

This solution provides scalability, security, and efficient management for the organization's IT infrastructure.

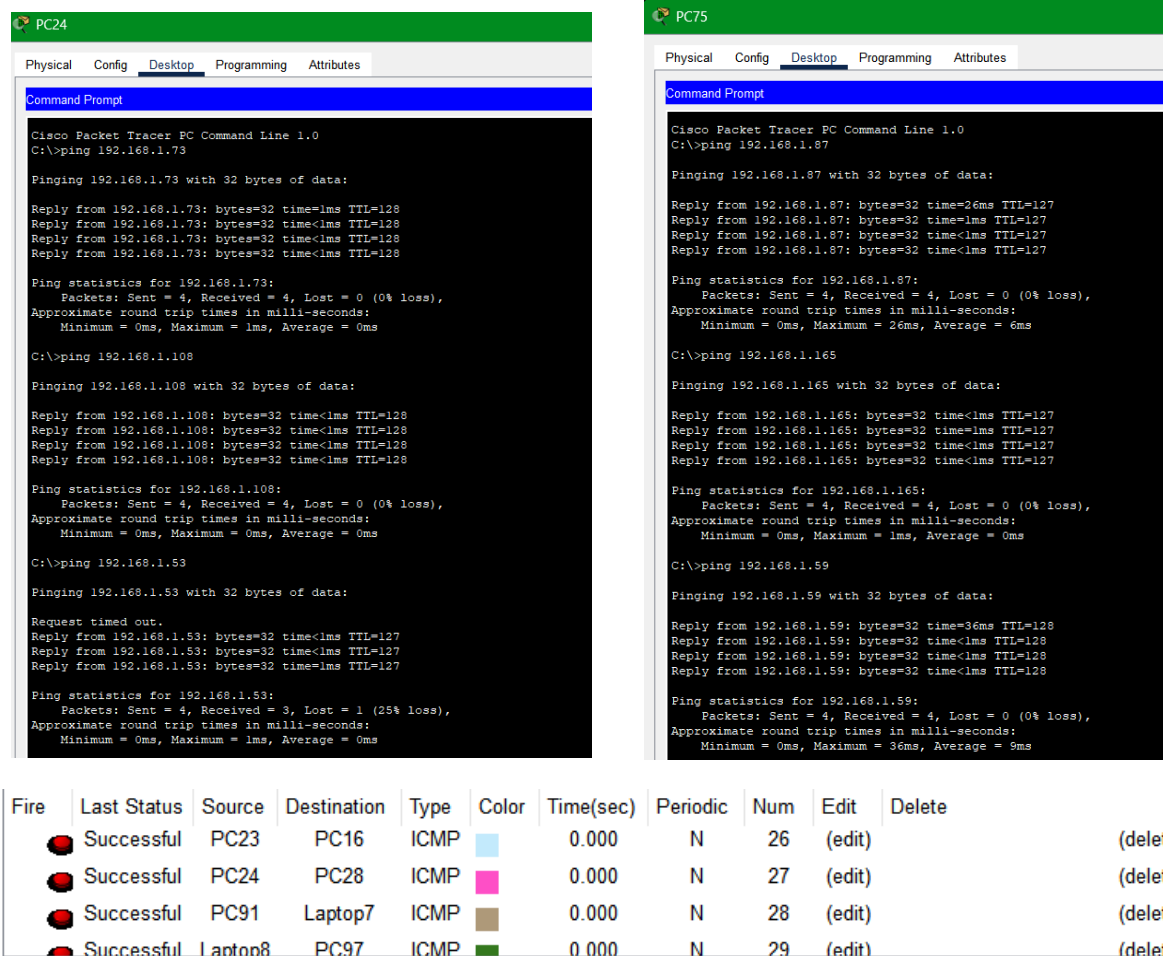


Figure 4: Successful Inter-VLAN Communication